A SEA OF XPUS: OCTAVALENT CELLS AND LANE EFFICIENCY

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DÆ RESEARCH

We refine the analysis of XPU-based interconnects by considering an *octavalent* cell design: each XPU exposes 8 symmetric ports arranged as "king moves" (north, south, east, west, northeast, northwest, southeast, southwest) in a 2D lattice. This differs from the earlier 8-dimensional mesh assumption, and alters the effective hop metric from Manhattan (ℓ_1) to Chebyshev (ℓ_∞). We compare Total Lane Capacity (TLC), Effective Link Bandwidth (ELB), and Packet Energy Intensity (PEI) for Clos vs. Octavalent Mesh, adding references from graph theory to ground the metrics in the literature.

TERMINOLOGY AND METRICS

Lane. A 100 Gb/s physical link.

TLC. *Total Lane Capacity*. Sum of endpoint and transit lanes.

EPLC. *Endpoint Lane Capacity*. Lanes attached to compute endpoints.

NELR. *Non-Endpoint Lane Ratio*. Fraction of lanes not attached to endpoints.

h. Average hop count.

ELB. *Effective Link Bandwidth*. TLC$/h$, measured in lane-equivalents.

PEI. *Packet Energy Intensity*. $h \times E_{\text{hop}}$.

TLC: Total Lane Capacity. See Harary (1969) *Graph Theory*.

EPLC: Endpoint Lane Capacity. Diestel (2017) *Graph Theory*.

h: Expected graph distance. Cf. Bollobás (1998) *Modern Graph Theory*.

ARCHITECTURAL MODELS

CLOS FABRIC

- Servers: 8 lanes each (2 ports × 4 lanes).
- Balanced, non-oversubscribed: NELR ≈ 50%.
- Average hop count $h_{\text{Clos}} \approx 4$.
- TLC ≈ $16N$ lanes; ELB ≈ $4N$ lanes.

OCTAVALENT XPU MESH

- Each XPU: 8 ports in king-move directions.
- Graph metric: Chebyshev (ℓ_∞ distance).
- On an $L \times L$ torus ($N = L^2$), average distance per dimension $\approx L/3$.

Chebyshev metric: see Chung (1997), *Spectral Graph Theory*.

- Thus,

$$h_{\text{mesh}} \approx \frac{1}{3}\sqrt{N}.$$

- TLC $= 8N$ lanes; ELB $\approx 8N/h_{\text{mesh}}$ lanes.

QUANTITATIVE COMPARISON

Machines	h_{mesh}	TLC_{Clos}	ELB_{Clos}	TLC_{Mesh}	ELB_{Mesh}
10^2	3.33	1,600	400	800	240
10^3	10.5	16,000	4,000	8,000	762
10^4	33.3	160,000	40,000	80,000	2,400
10^6	333.3	16,000,000	4,000,000	8,000,000	24,000

Table 1: Clos vs. Octavalent Mesh (king-move). All values in lanes; ELB = Effective Link Bandwidth.

INTERPRETATION

CLOS

Uniform-random traffic benefits from:

- Short hop count ($h = 4$).
- Ample transit capacity (NELR $\approx$ 50%).

OCTAVALENT MESH

Advantages:

- Fabric tax (FT) = 0; all lanes terminate at XPUs.
- Energy-efficient if $E_{\text{hop,XPU}} \ll E_{\text{hop,Clos}}$.
- With locality-aware scheduling, effective h can be reduced to near-neighbor scales ($h \leq 2$), dramatically improving ELB and lowering PEI.

REFERENCES

- Harary, F. (1969). *Graph Theory*. Addison-Wesley.
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